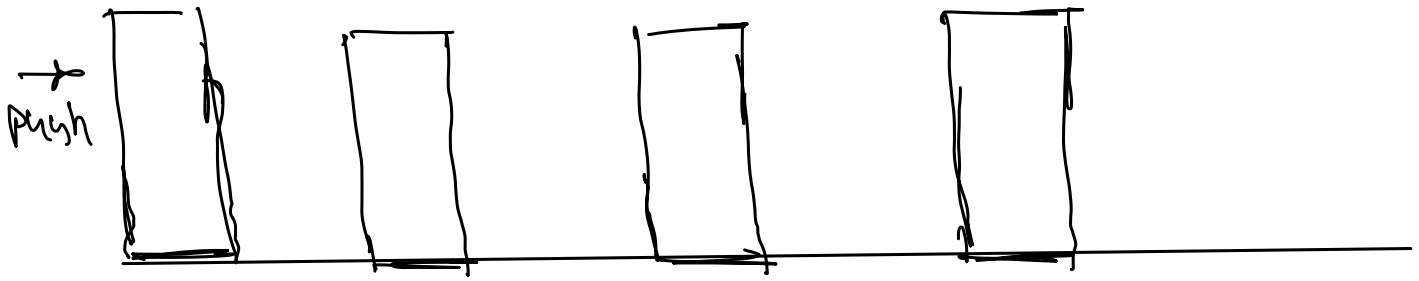


Induction: (Jay Cummings)

Domino:



- 1st domino will fall into the next one
- every domino when it's knocked falls onto the next one,
- domino no: n will fall yes or no!

it depends on whether domino $(n-1)$ fell or not,

eventually this question boils down whether the 1st domino fell or not,

Now, let's look at an example:

• 1st case: $1 = 1^2$

2nd case: $1 + 3 = 4 = 2^2$

3rd case: $1 + 3 + 5 = 9 = 3^2$

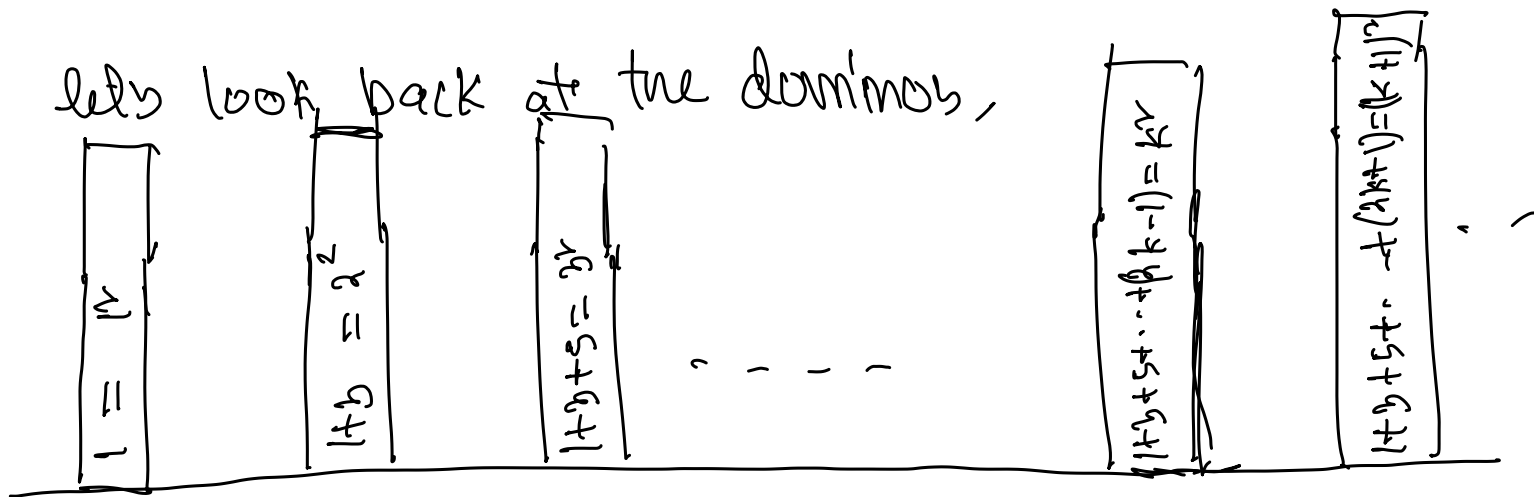
⋮

k^{th} case: $1 + 3 + 5 + 7 + \dots + (2k-1) = k^2$

sum of first n odd integers are n^2

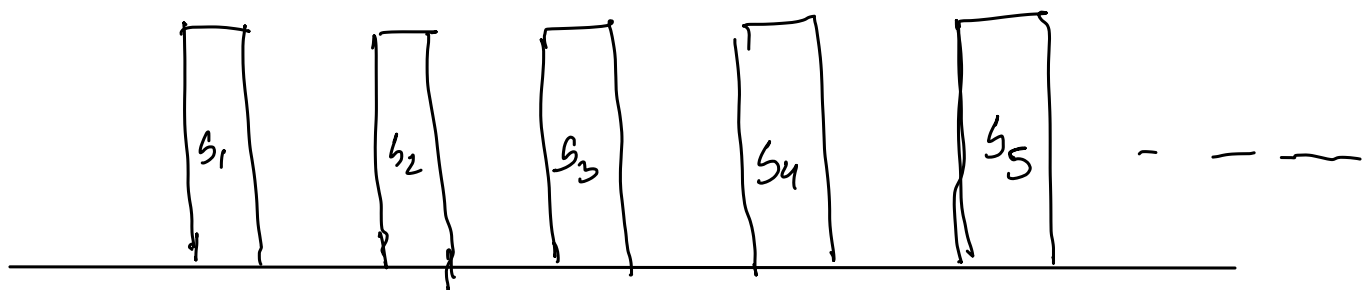
⇒ Proving individual case by case is not possible,

lets look back at the dominos,



there are all sequence of statements

Suppose, we are given a sequence of statements, $s_1, s_2, s_3, s_4, \dots$



• Suppose s_1 is true

• for each, $k \in \mathbb{N}$, if s_k is true then, s_{k+1} is true

Then s_n is true for every $n \in \mathbb{N}$

(Principle of induction)

Proposition: $s_1, s_2, s_3, \dots$ all are true

Proof:

(i) Base case $\langle s_1 \text{ is true} \rangle$

(ii) inductive hypothesis: Assume s_k is true

(iii) Induction step: if s_k then s_{k+1}

(iv) Conclusion: by induction, all the s_n are true

example: $1 + 3 + 5 + \dots + (2k-1) = k^2$

Proof:

Base case: For $k=1$ the statement holds,

since, $1 = 1^2$ [s_1 is true]

Inductive hypothesis:

we assume, $k=n$ the statement is true,

then, $1 + 3 + \dots + 2n-1 = n^2$ (s_n is true assumed)

Induction step: ($s_n \rightarrow s_{n+1}$)

$$\underbrace{1 + 3 + 5 + \dots + (2n-1)}_{n^2} + (2n+1)$$

$$= n^2 + 2n + 1 = (n+1)^2$$

$\therefore s_{n+1}$ is true

Conclusion: s_n is true, $\forall n \in \mathbb{N}$

Proposition:

For every $n \in \mathbb{N}$

$$1+2+\dots+n = \frac{n(n+1)}{2}$$

(Proof by induction)

Direct Proof:

$$s_n = 1+2+\dots+n$$

$$s_n = n+(n-1)+\dots+1$$

$$2s_n = (n+1) + (n+1) + \dots + (n+1)$$

$$s_n = \frac{n \cdot (n+1)}{2}$$

Proposition: s_n be the sum of first n natural number, then for any $n \in \mathbb{N}$

$$P_n: s_n + s_{n+1} = (n+1)^2$$

Proof:

Base case: $n=1$,

$$s_1 = 1 \quad s_2 = 1+2 = 3$$

$$\therefore s_1 + s_2 = 1+3 = 4 = (1+1)^2$$

∴ For $n=1$ the statement, P_1 is true,

Inductive hypothesis:

We assume the statement is true for $n=k$

$$\therefore S_k + S_{k+1} = (k+1)^2$$

Induction step: ($P_k \rightarrow P_{k+1}$)

$$\begin{aligned} & S_{k+1} + S_{k+2} \\ &= \overbrace{(1+2+3+\dots+k+(k+1))}^{S_k} + \overbrace{(1+2+3+\dots+(k+1)+(k+2))}^{S_{k+1}} \\ &= S_k + S_{k+1} + (k+1) + (k+2) \\ &= (k+1)^2 + (k+1) + (k+2) \\ &= (k+1)^2 + (k+1) + (k+1) + 1 \\ &= (k+1)^2 + 2(k+1) + 1 \\ &= (k+1+1)^2 \end{aligned}$$

Conclusion: P_n is true, for every $n \in \mathbb{N}$

Direct Proof:



$$\begin{aligned}S_n + S_{n+1} &= \frac{n(n+1)}{2} + \frac{(n+1)(n+2)}{2} \\&= \frac{(n+1)}{2} (n + n+2) \\&= \frac{(n+1) \cdot 2 \cdot (n+1)}{2} = (n+1)^2\end{aligned}$$

Proposition: For every $n \in \mathbb{N}$, the product of first n odd natural numbers equals $\frac{(2n)!}{2^n n!}$.

$$S_n: 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1) = \frac{(2n)!}{2^n n!}$$

Proof:

Base case:

$n=1$ statement is true,

$$1 = \frac{(2 \cdot 1)!}{2^1 \cdot 1} = \frac{2}{2 \cdot 1} = 1$$

(s_1 is true)

Inductive hypothesis:

For $n=k$ the statement is true,

$$1 \cdot 3 \cdot 5 \cdot \dots \cdot (2k-1) = \frac{(2k)!}{2^k k!}$$

Induction step: ($s_k \rightarrow s_{k+1}$)

$$\begin{aligned} & 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2k-1) \cdot (2k+1) \qquad (2(k+1))! \\ &= \frac{(2k)!}{2^k \cdot k!} \cdot (2k+1) \cdot (2k+2) \\ &= \frac{(2k+1)! \cdot (2k+2)}{2^k \cdot k! \cdot (2k+2)} = \frac{(2k+2)!}{2^{k+1} (k+1)!} \end{aligned}$$

Conclusion: s_n is true, for every $n \in \mathbb{N}$
(direct proof, HW)

Proposition:

$$1 + 2 + 4 + 8 + \dots + 2^n = 2^{n+1} - 1$$

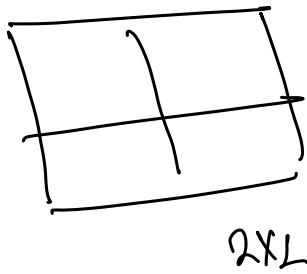
Prove by induction & direct proof

Proposition:

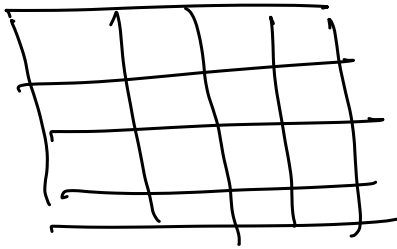
$$1 + a + a^2 + \dots + a^n = \frac{(a^{n+1} - 1)}{(a - 1)}$$

Prove by induction & direct proof,

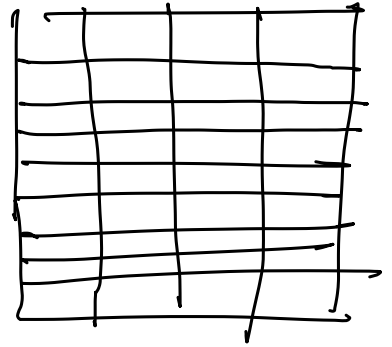
Tiling Problem:



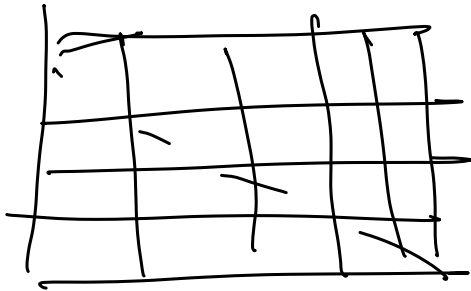
2x1



4x4

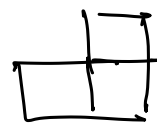
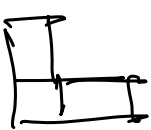


8x8



2^n x 2^n

use:



to cover the grids

Can I cover it?

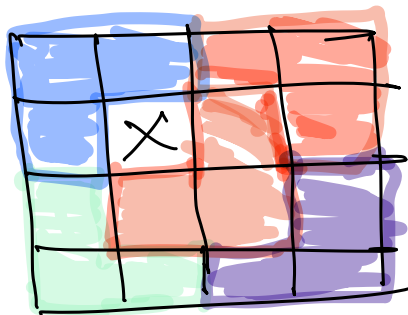
No, since $3 \nmid 2^n \times 2^n \Rightarrow 3 \nmid 4^n$

But, $3 \mid (4^n - 1)$

No matter

where you

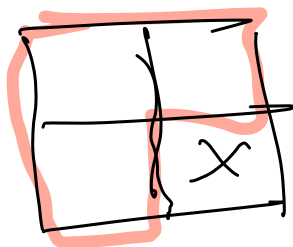
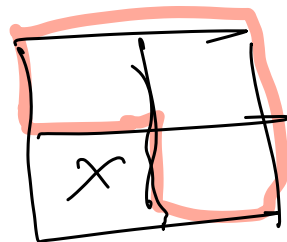
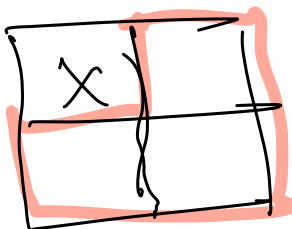
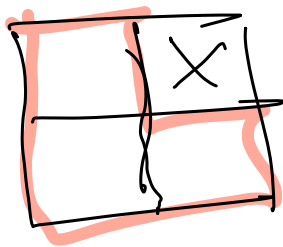
put the X you
will always find
a covering,



Proposition: For every $n \in \mathbb{N}$ if any one square
is removed, from a $2^n \times 2^n$ chess board the
result can be perfectly covered with L-shaped tiles

Proof: when, $n=1$

grid size $= 2^1 \times 2^1$



possible:

Inductive hypothesis:

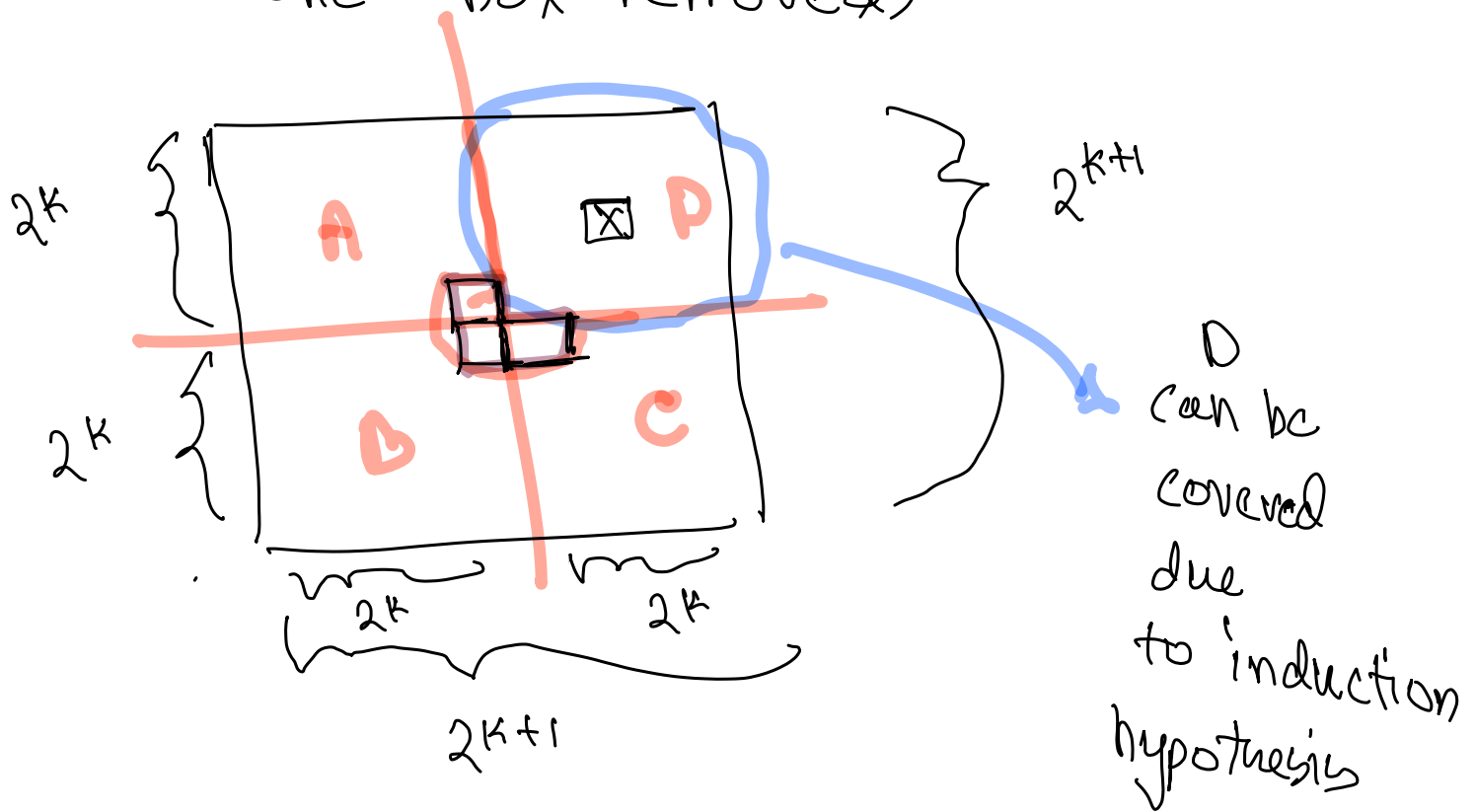
let, $k \in \mathbb{N}$ such that, $k \geq 2$ we assume that,

if any square is removed from $2^k \times 2^k$ it

can be perfectly covered,

Inductive step:

assume a $2^{k+1} \times 2^{k+1}$ chess board
with one box removed,



A, B, C is filled in such a way
that only the above shown grids
remain empty,

Now fill up the empty grid with a single
tile

Conclusion: